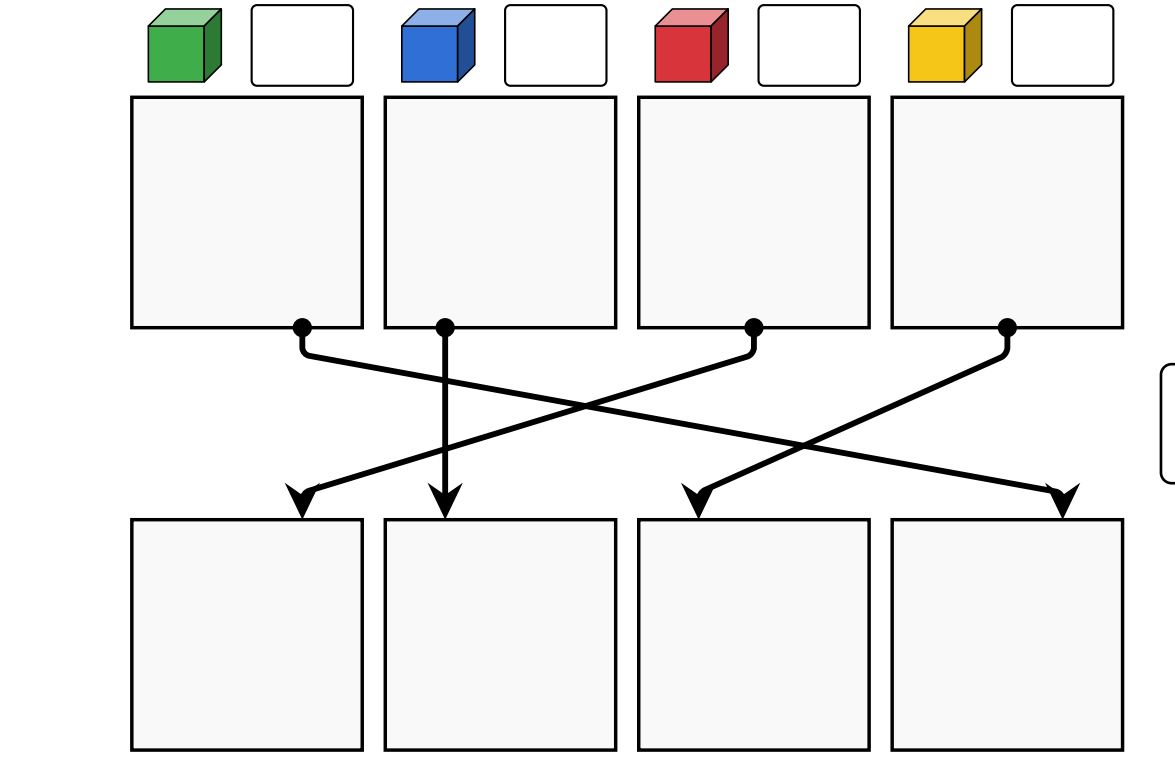
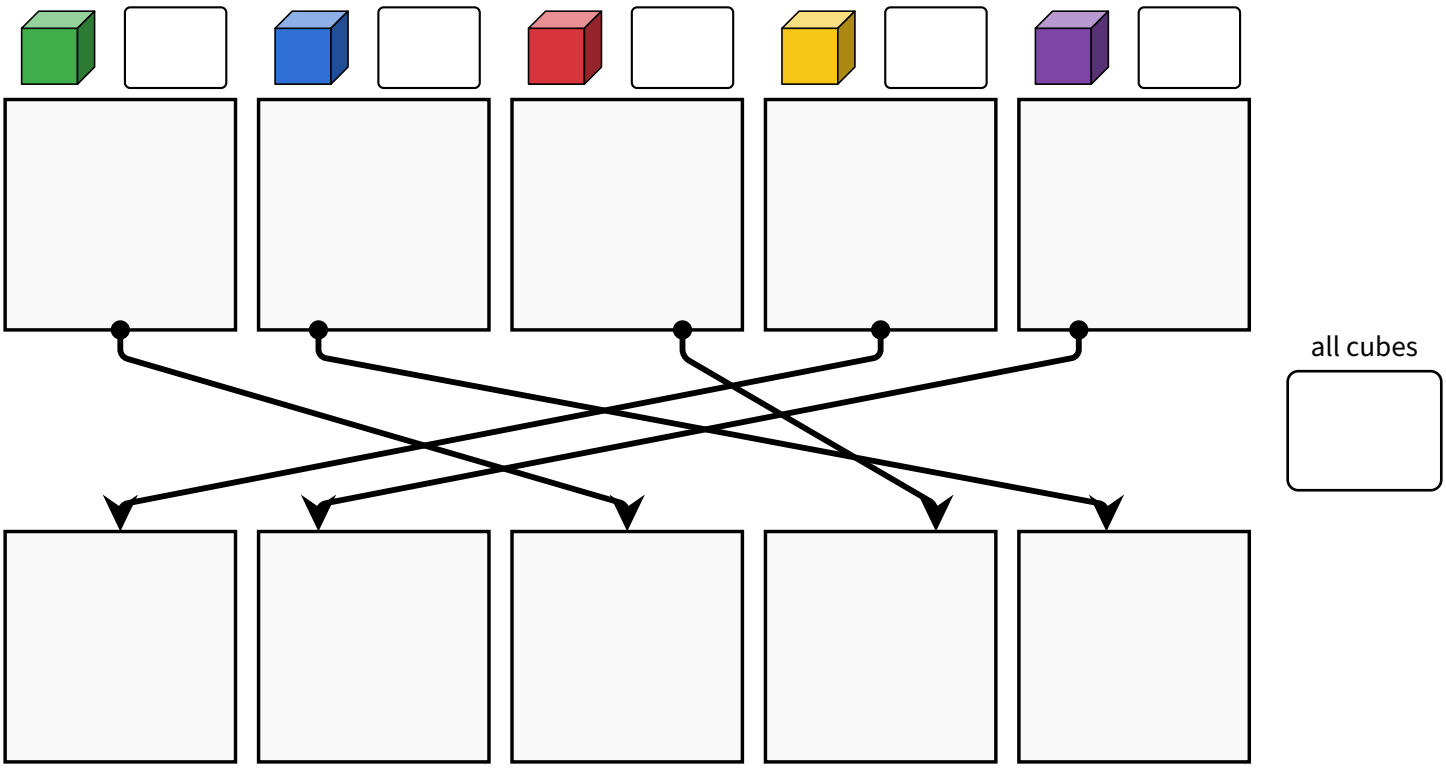
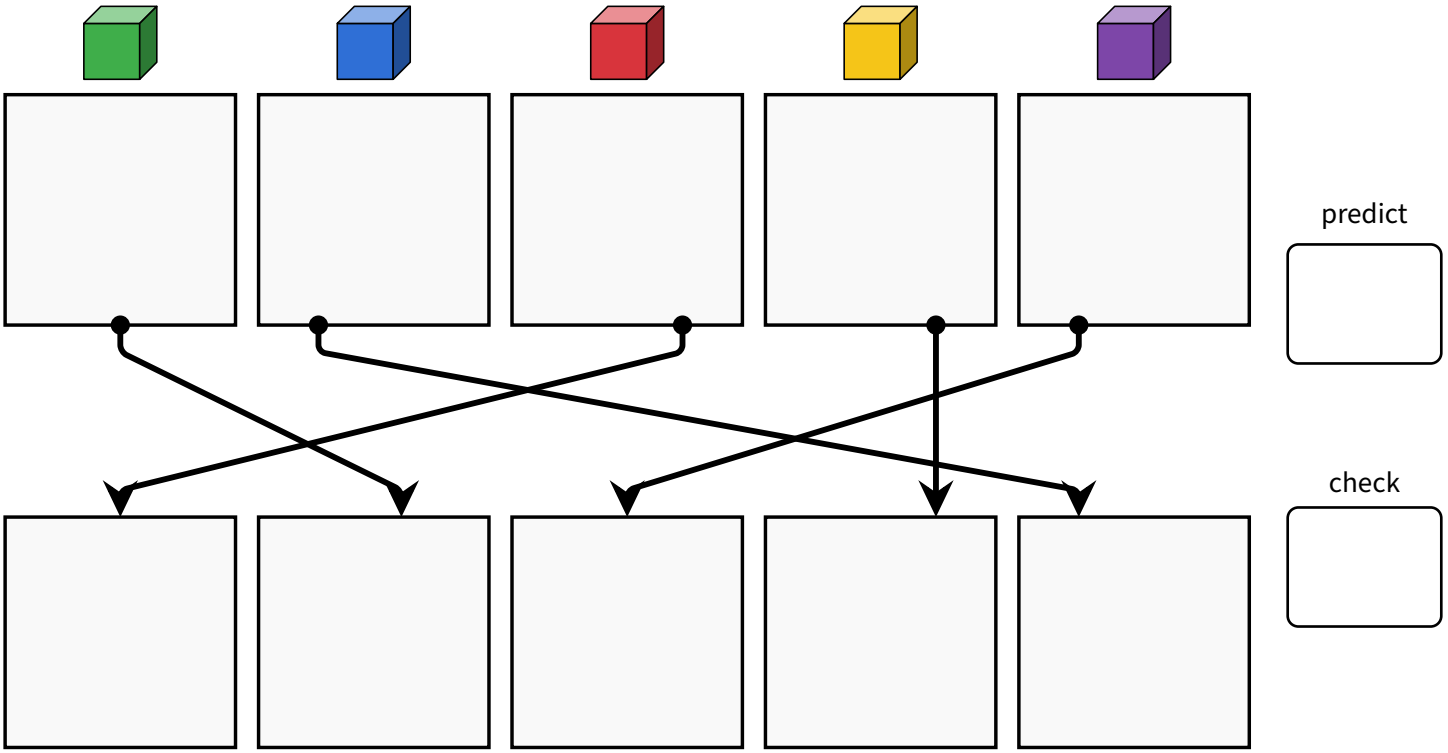
One turn: move every cube down its own arrow, then slide the whole bottom row straight up into the top row. When you draw a machine, give every top slot one arrow going out and every bottom slot one arrow coming in. A cube is home when it is back under its own picture.

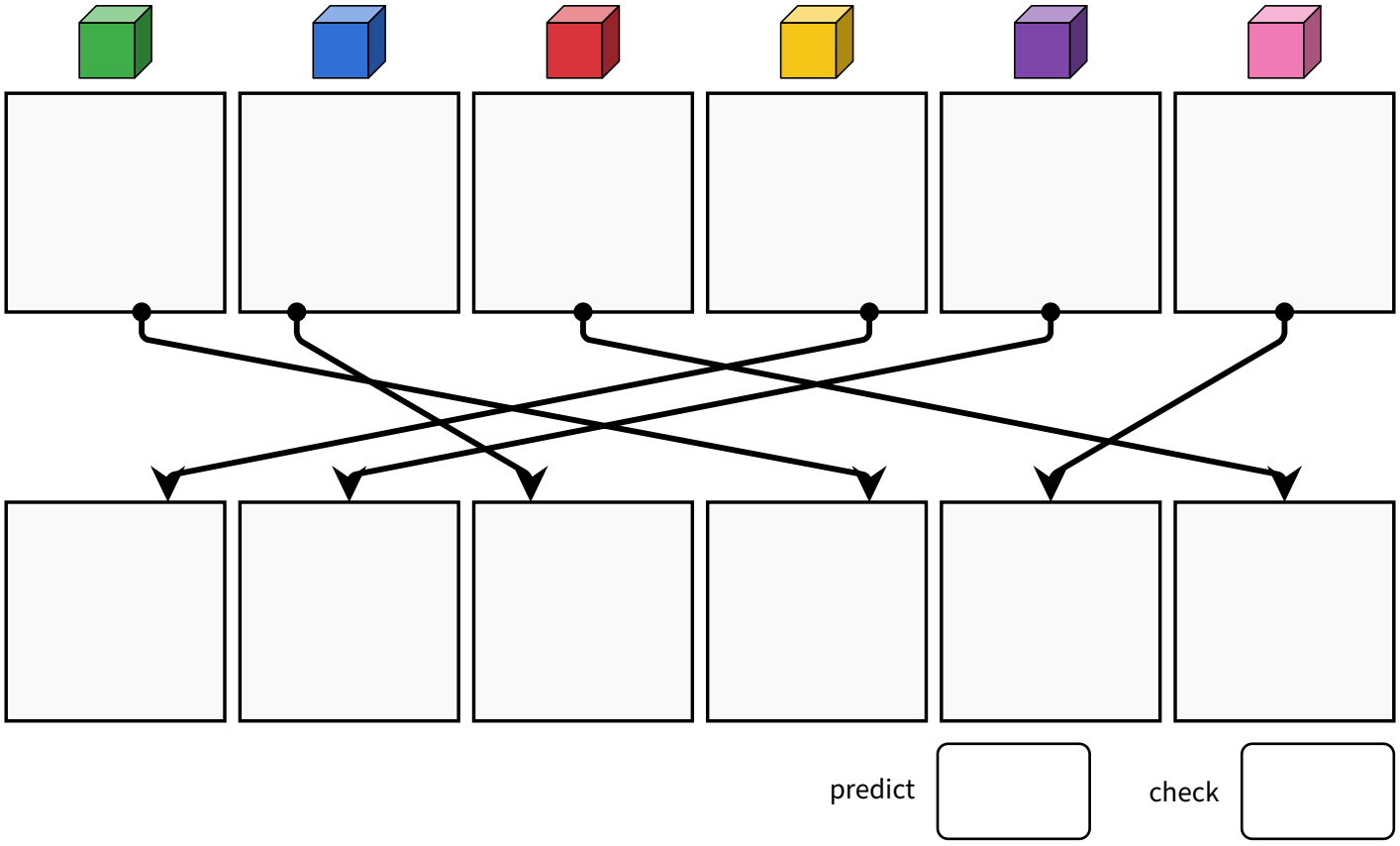
Problem 1: Put each cube under its picture and run each machine. For each cube, find the first turn after which it is back home. Then find the first turn after which all the cubes are back home at the same time.

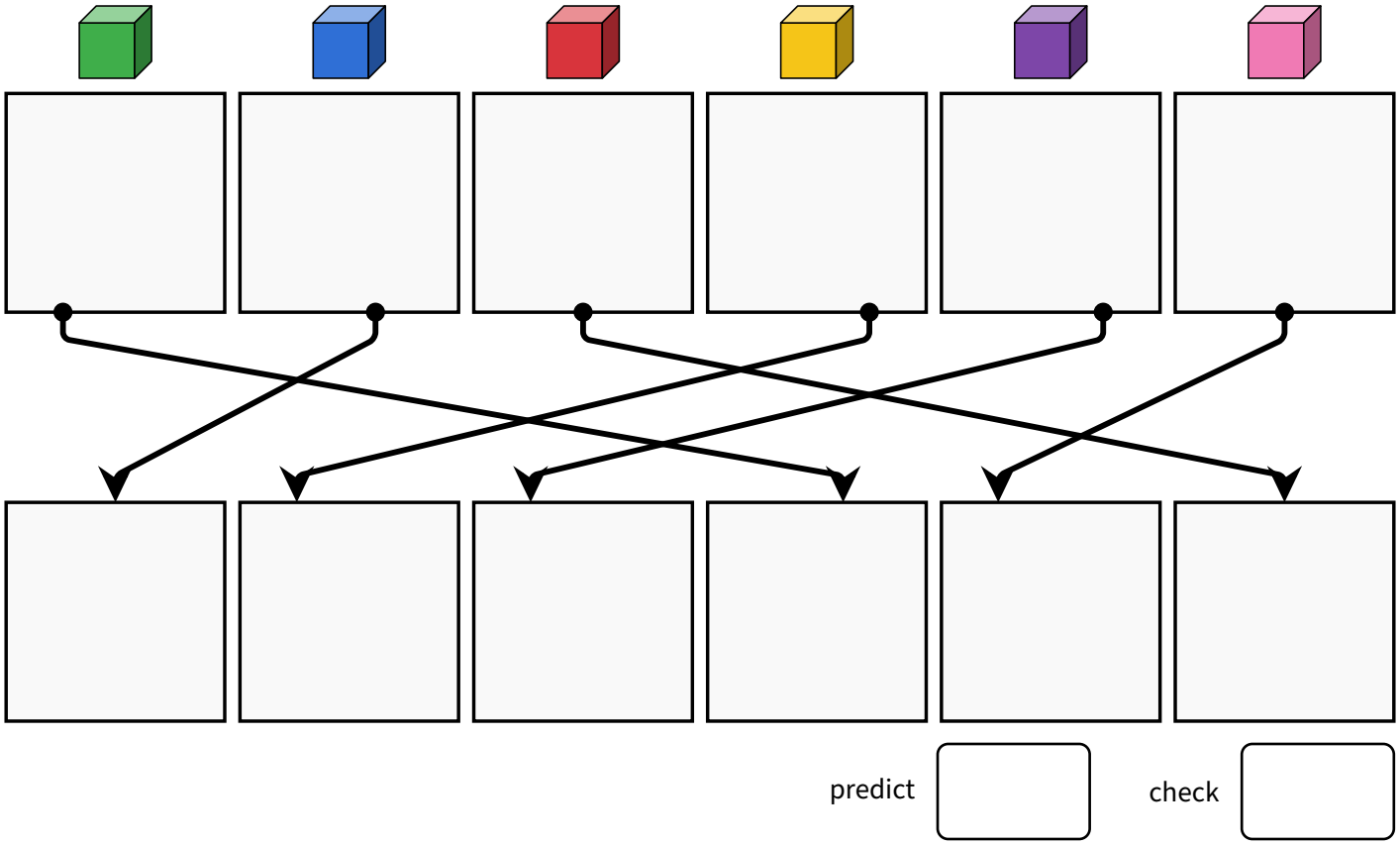


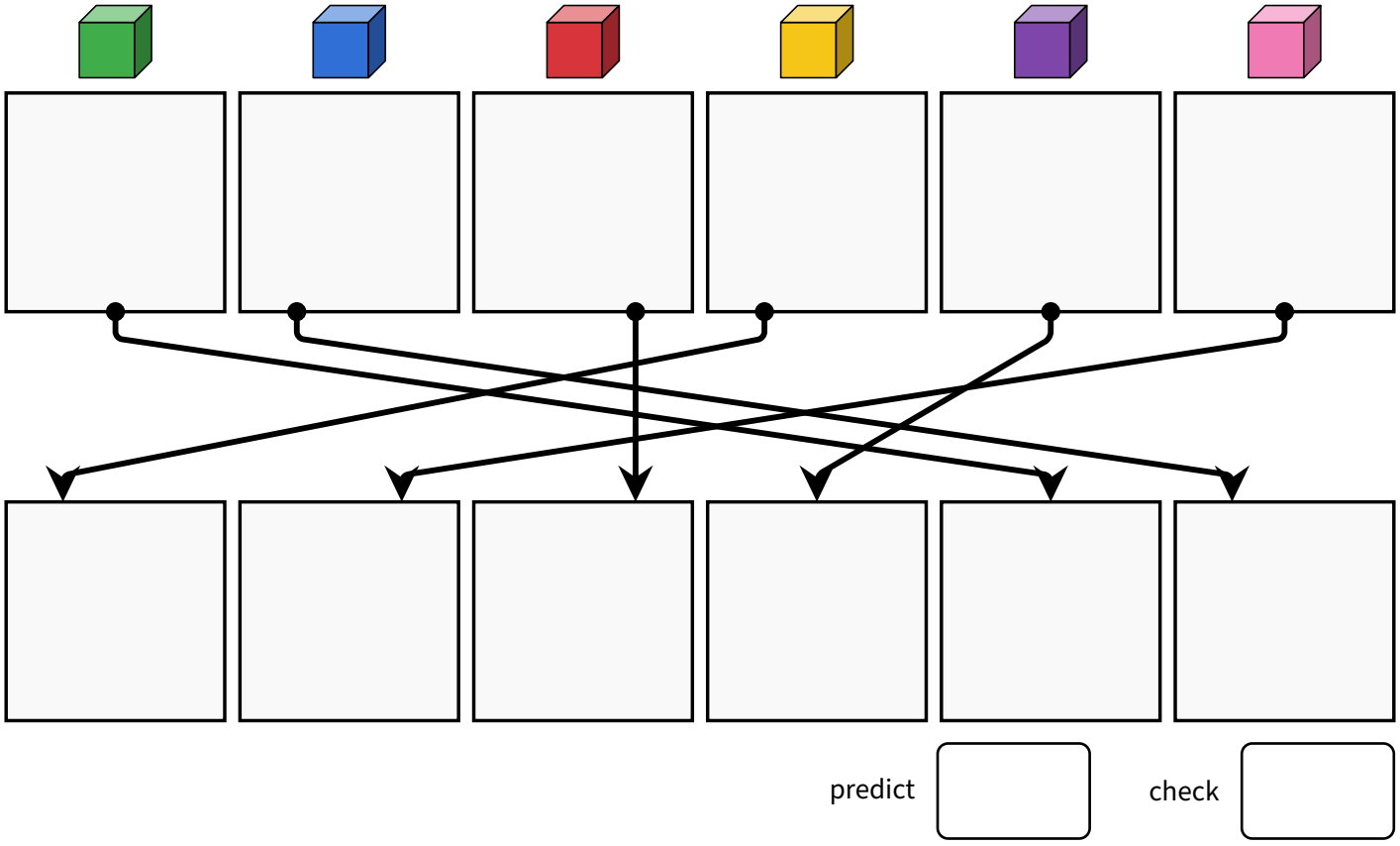


Problem 2: A cube's own slot and the slots it visits on its way home make a loop. Without moving any cubes, colour the top slots so that each loop has its own colour, and predict how many turns the machine takes to bring all the cubes home. Then run it to check. Do this for each of the four machines.



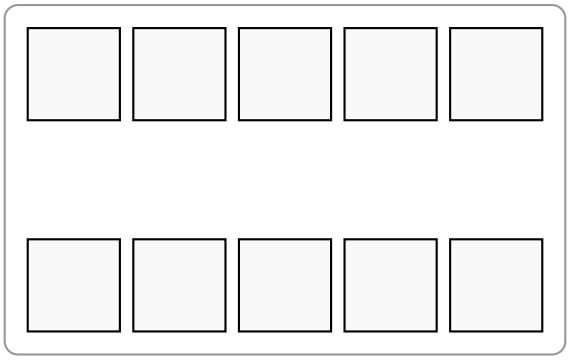



predict
check

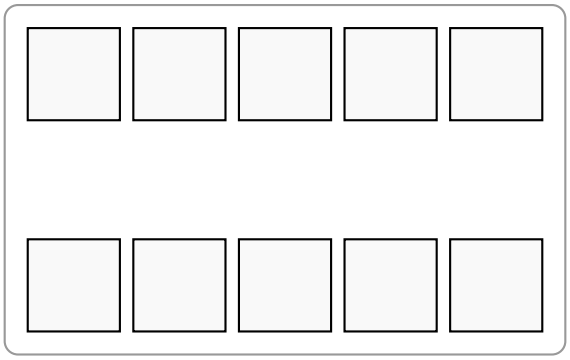

predict
check

Problem 3: For each number of turns from 1 to 6, draw a machine with 5 slots that takes exactly that many turns to bring all the cubes home.

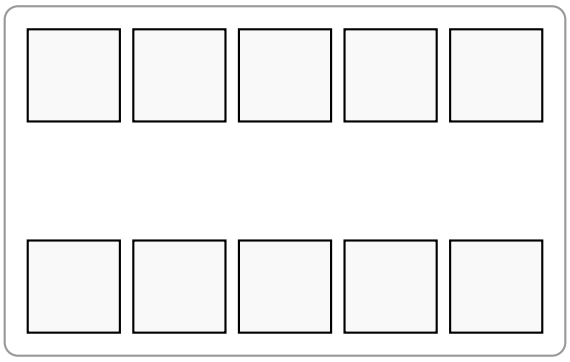
1 turn



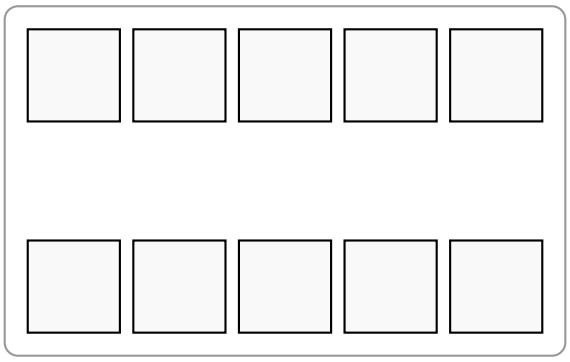
2 turns



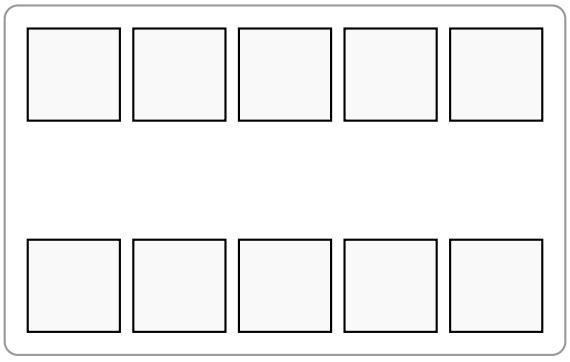
3 turns



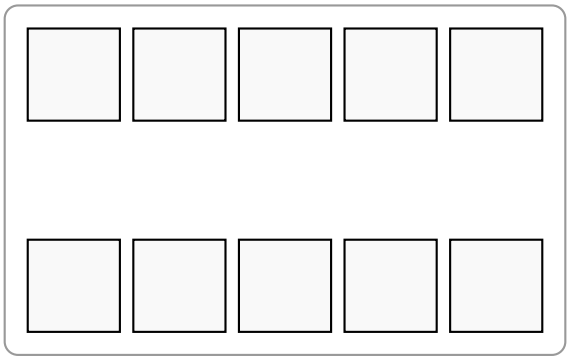
4 turns



5 turns



6 turns



Problem 4: Put each cube under its picture on a machine with arrows and do one turn. Move the row of cubes to the empty machine below it without changing its order. Draw arrows on the empty machine so that one turn of it puts every cube home. Do this for both machines with arrows.

